Nicolò Girardello

MSc Student in Control Systems Engineering

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Portfolio: girardellon.github.io
Languages: Italian, English, Spanish

EDUCATION

Sep 2026 – Present **MSc in Control Systems Engineering** — University of Padua, Italy
Focus on control systems, robotics, autonomous systems and artificial intelligence, with emphasis on robot control, reinforcement learning, computer vision and deep learning.

Sep 2025 – Jan 2026 **Erasmus+ Exchange Program** — Universitat Politècnica de Catalunya, Spain
Robotics and control engineering. Coursework and practical experience in mobile robotics, power electronics and MATLAB-based engineering applications.

Sep 2023 – Jul 2026 **BSc in Automation and Systems Engineering** — University of Padua, Italy
Focus on control systems, robotics and machine learning, with strong foundations in mathematics, electronics and computer engineering.
Thesis: *Tactile-to-Visual Representation Learning for Vision-Language-Action Models*

Jul 2022 **Erasmus+ Exchange Program** — Heraklion, Greece
Professional internship at Dimitriadis Marine Service, gaining practical experience in an international technical environment.

Sep 2018 – Jun 2023 **Technical High School Diploma** — ITIS Alessandro Rossi, Italy
Specialisation in electronics, electrical engineering and automation. Design of electrical and electronic systems and industrial automation.

EXPERIENCE

Nov 2023 – Present **Technical Office Specialist** — MEP Solution Srl, Italy
Contributed to the design and technical documentation of electrical and photovoltaic systems, including technical analysis and cost estimation.

Feb 2026 **Robotics Research Intern** — RoboTac Lab at BMW Group, Germany
Contributed to the development and integration of ROS 2-based robotic systems, with a focus on motion planning and experimental validation in a research environment.

ACADEMIC EXPERIENCE & PROJECTS

Full reports and videos at girardellon.github.io

Jul 2026 **Tactile-to-Visual Representation Learning for VLA Models**
BSc Thesis Project, University of Padua
Maps DIGIT tactile embeddings into the frozen visual latent space of a VLA model.

Jun 2026 **Redundancy Resolution: Kinematic Control for Franka Panda**
"Introduction to Robotics", University of Padua
CLIK and null-space projection to track a trajectory while avoiding joint limits.

Jan 2026 **Line Following & Evolutionary Obstacle Avoidance**
"Sensors, Actuators & Microcontrollers", UPC Barcelona
PID line following and genetic-algorithm-evolved neural obstacle avoidance.

Dec 2025 **Multi-Robot Warehouse Optimization Simulator**
"MATLAB and its application in engineering", UPC Barcelona
A* and Dijkstra path planning, collision avoidance and energy-aware task allocation.

Nov 2022 **ChessBot: Automated 3-Axis Chess-Playing Robot**
Personal Project
Cartesian robot linking a PLC with a Python chess engine.

May 2022 **BerryBot: Automated Strawberry Harvesting System**
Siemens Automation Olympics 2022, 1st place regional
Vision-guided robotic arm that identifies and picks ripe strawberries.

May 2021 **Smart Warehouse: Automated Material Dispensing**
SCRUM project, ITIS Alessandro Rossi
Carousel storage and dosing system with an integrated Archimedes screw.